

Diagonal lift of the interpolant residual

$n_-(W_Y - K_{YY}^{-1}) = \#\{\lambda(W^{1/2}KW^{1/2}) < 1\}$; the bound ≤ 1 is not a $d_{\min}$ theorem

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Abstract

R439 left $D_0 = W_Y - K_{YY}^{-1}$ and, at kernel dimension 0, the pair $\Delta D_0 \Delta = L_{-\tilde{m}} + \text{diag}(w_Y(P'_Y)^2)$. This round asks the one lemma **DIAG_LIFTS**: $n_-(D_0) \leq 1$ on the selected sequence, equivalently whether the PD diagonal lifts all but at most one negative direction of $L_{-\tilde{m}}$. $L_{\tilde{m}}$ is the dressed Cauchy Gram $C \text{diag}(w\pi_Y(X)^2) C^T$, hence always PSD; X -between- Y interlacing does *not* change this inertia. The lift count is exact: $n_-(W_Y - K^{-1}) = \#\{\lambda(W^{1/2}KW^{1/2}) < 1\}$ (and at $k_{\dim} = 0$, $\#\{\lambda(D^{-1/2}LD^{-1/2}) < -1\}$). The implication $d_{\min} = \min_i w_i K_{ii} > 1 \Rightarrow n_- \leq 1$ is false already on the six-node $\mathbb{Q}$ -window. The selected bound remains a census. Ausgang REDUZIERT. No RH claim.

Status. Lemmas 2–3 are proved (finite identities). Proposition 1 is a named refutation. Proposition 2 is a named refutation. Propositions 3–4 are finite machine checks. Ausgang REDUZIERT. No L^* claim. No $R^\dagger$ claim. No RH claim.

1 The lemma

Lemma 1 (DIAG_LIFTS, two forms). *On a window let $K = K_{YY}$ be the RKHS kernel of $P_{<d_0}$ on the X -measure and $W_Y = \text{diag}(u^\vee|_Y)$. The interpolant residual is $D_0 = W_Y - K^{-1}$. **DIAG_LIFTS** asks $\text{ind}_-(D_0) \leq 1$ on the selected sequence. At $k_{\dim} = 0$ this is the pair statement: the PD diagonal $\text{diag}(w_Y(P'_Y)^2)$ lifts all but at most one negative direction of $L_{-\tilde{m}}$.*

2 Loewner signature is a Gram

Write $\tilde{m}(z) = \sum_a u^\vee(x_a) P_Y(x_a)^2 / (x_a - z)$. The Loewner matrix at Y is the Cauchy Gram

$$(L_{\tilde{m}})_{ij} = \sum_a \frac{u^\vee(x_a) P_Y(x_a)^2}{(x_a - y_i)(x_a - y_j)}. \quad (1)$$

Lemma 2 (Gram). $L_{\tilde{m}} = C \text{diag}(w\pi_Y(X)^2) C^T$ exactly over $\mathbb{Q}$ on both r431 toys. Hence $L_{\tilde{m}}$ is PSD of rank $\min(|X|, |Y|)$ and $L_{-\tilde{m}}$ is NSD of the same rank. On the five-atom toy the inertia of $L_{-\tilde{m}}$ is $(0, 2, 0)$; on the six-node toy $(0, 3, 0)$.

Proposition 1 (Interlacing does not set L -inertia). *The five-atom toy has no X -node in $(\min Y, \max Y)$; the six-node toy has one. Both $L_{-\tilde{m}}$ are fully ND. Classical “poles between nodes change Loewner inertia” does not apply to this source $\tilde{m}$: the Gram (1) is PSD regardless of interlacing. Fold geometry is not an L -inertia statement.*

Dropping the diagonal (the drop- D mutant) restores the full ND signature: $\text{ind}_-(L_{-\tilde{m}}) = |Y|$ on both toys.

3 The lift count

At $k_{\text{dim}} = 0$, $\Delta D_0 \Delta = L_{-\tilde{m}} + D$ with $D = \text{diag}(w_Y(P'_Y)^2) \succ 0$. Sylvester gives $\text{ind}_-(D_0) = \text{ind}_-(L + D) = \#\{\lambda(D^{-1/2}LD^{-1/2}) < -1\}$. On live windows $k_{\text{dim}} \gg 0$ the pair identity fails (relative residual 0.4–400); the Cauchy– π evaluation of L overflows. The general identity does not need the pair:

Lemma 3 (Count). *If $W_Y \succ 0$ and $K \succ 0$ then*

$$\text{ind}_-(W_Y - K^{-1}) = \#\{\lambda(W_Y^{1/2}KW_Y^{1/2}) < 1\}. \quad (2)$$

At $k_{\text{dim}} = 0$ this equals the -1 -count of the normalised Loewner. On both $\mathbb{Q}$ -toys the two counts agree with $\text{ind}_-(D_0)$ (five-atom: 1; six-node: 2).

The -1 gap of the pair form is a $k_{\text{dim}} = 0$ object. The live selected coordinate is the gap of the dual-weighted CD kernel about 1.

4 Christoffel and $d_{\min}$

Cauchy–Schwarz on K gives $(K^{-1})_{ii}K_{ii} \geq 1$, exactly 5592721/252000 on the five-atom (strict). The r408 separator is the diagonal of the weighted kernel, $d_{\min} = \min_i w_i K_{ii}$.

Proposition 2 ($d_{\min} > 1 \not\Rightarrow n_- \leq 1$). *On the six-node $\mathbb{Q}$ -window $d_{\min} = 210157/57600 > 1$ and $\text{ind}_-(D_0) = 2$. The implication is false as a theorem. It remains a MAIN/adversary census: selected and χ windows in the sealed sample have $d_{\min} > 1$; permute/jitter/scramble have $d_{\min} < 1$ and many slips.*

$\text{diag}(D_0)$ is not the carrier: MAIN w_9 has no negative diagonal entry and $\text{ind}_- = 1$; permute has 32 negative diagonal entries and $\text{ind}_- = 20$.

5 Windows

Proposition 3 (Selected / kz52). *On $w_9, k_z15, k_z18, k_z20, k_z52$ the count (2) equals 1. Gaps of $W^{1/2}KW^{1/2}$ about 1: w_9 0.857–1.00018; k_z15 0.816; k_z18 0.640; k_z52 0.781 ($d_{\min} = 1.272$). The r438 razor $\lambda_2(C) - 1 = 2.2 \cdot 10^{-7}$ is a different depth; in this coordinate k_z52 is one slip with a comfortable gap. The tight selected slip in the sealed sample is k_z22 (0.99981).*

Proposition 4 (Kills / χ). Λ -permute: 20 slips, $d_{\min} = 0.148$. Jitter: 19 / 0.628. Scramble: 21 / 0.367. No gap at 1. χ_3 -9: 0 slips, $\lambda_{\min} = 1.00088$ (PD, world-separating). χ_3 -15: one slip at 0.99816. Death is the side of the 1-threshold, consistent with r440's collar side.

6 What remains

Remark 1 (Remainder). DIAG_LIFTS on Selected is equivalent to: the dual-weighted CD kernel $W_Y^{1/2}K_{YY}W_Y^{1/2}$ has at most one eigenvalue below 1. This is not an interlacing count of X versus Y , not a $d_{\min}$ theorem, and not a sign count of $\text{diag}(D_0)$. The named remainder is off-diagonal control of K (how far the weighted kernel can dip twice below 1 while every diagonal exceeds 1). The six-node toy shows that this can happen off Selected. The selected bound is a census, not a proof. It is not a theorem of the Riemann hypothesis.

Mincut unchanged (base 4 / refined 5). No L^* claim. No $R^\dagger$ claim. No RH claim.